AIAA 3201 · PROJECT 4 · PART 1

Initialization, not architecture

COLMAP vs. VGGT-1B vs. VGGT-Ω as 3D Gaussian Splatting initialization under a controlled dense view budget

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What we compared, and how

FOUR AXES, AS SPECIFIED IN THE BRIEF

Initialization runtime
seconds to a training-ready scene

Pose accuracy
ATE after Sim(3) alignment; relative rotation

3DGS convergence speed
test PSNR every 5 k iterations to 30 k

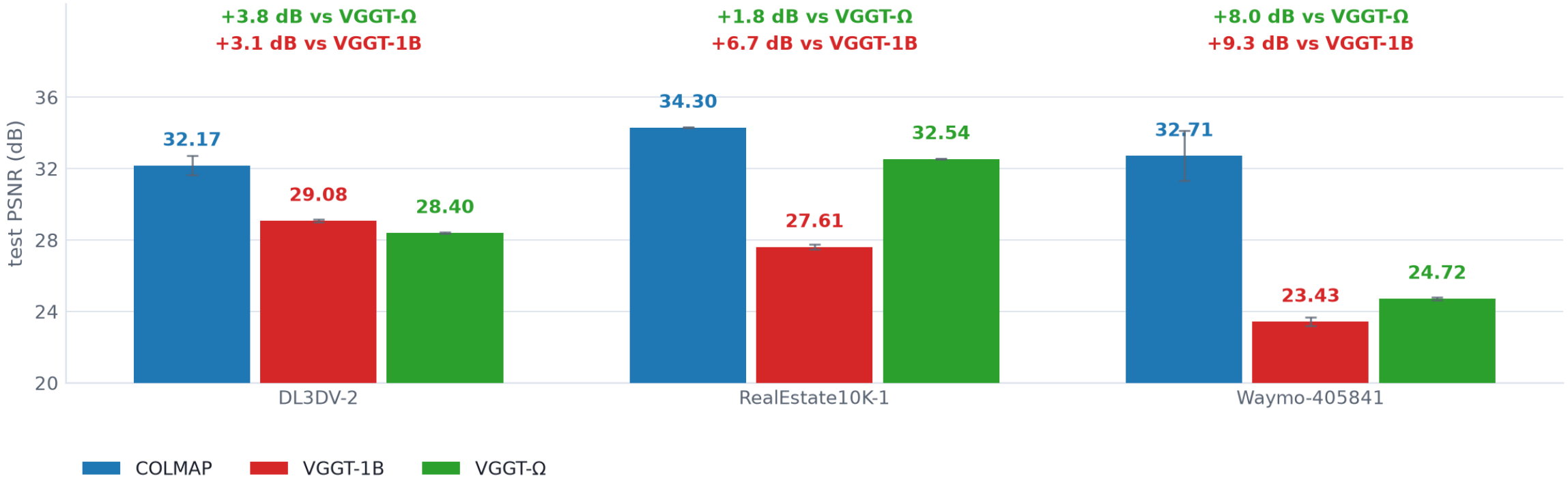
Final rendering quality
PSNR / SSIM / LPIPS on frozen test views

THREE MANDATORY SCENES

Scene	Type	Images	Test
DL3DV-2	outdoor	306	39
RealEstate10K-1	indoor	279	35
Waymo-405841	driving	199	25

- N = 160 training views — the dense regime; 8–17× denser than Part 2's sparse settings
- Splits frozen, llffhold = 8 — all three backends see the same images
- 30 000 iterations, densify to 15 000, reference hyper-parameters
- 2–3 seeds per configuration, with pairwise t-tests

COLMAP produces the best renders on all three scenes



y-axis truncated at 20 dB · error bars = run-to-run std (2–3 seeds)

Largest gap on Waymo-405841 (+9.3 dB). SSIM and LPIPS rank the backends identically, so the ordering is not an artefact of one metric.

The ordering is visible without metrics



Column headers give the scene-level mean test PSNR (mean over all test views, 2–3 seeds where available); corner labels give that single view's PSNR.
Note: the official VGGT exporter caps the point cloud at 100 000 points chosen by a random unseeded subsample (vggt/utils/helper.py::randomly_limit_trues), so VGGT-1B figures carry an export-level spread far larger than its training-seed spread — see the report's limitations.

WHAT TO LOOK AT

VGGT-1B, 21.4 dB

loses the fine structure of the red foliage and the paving joints.

VGGT-Ω, 28.3 dB

keeps the shape but drifts in colour — visibly greyer and colder.

COLMAP, 32.7 dB

closest to ground truth on all three rows.

ONE THING TO BE CAREFUL ABOUT

These renders use a different VGGT-1B export than the 29.1 dB in the table. The exporter subsamples the cloud with an unseeded **np.random.choice**, so that column moves by several dB between exports of the same model.

The mechanism is not pose accuracy

POINT CLOUD, DL3DV-2 (model_analyzer)

Model	Points	Mean track length
COLMAP	3 778	**8.19**
VGGT-1B	100 000	1.000
VGGT-Ω	150 000	1.000

TRACK LENGTH = 1.000 MEANS EVERY 3D POINT IS SEEN IN EXACTLY ONE IMAGE

— Both exporters emit per-frame depth unprojections, not multi-view triangulations.

— 3DGS therefore starts from mutually unconstrained samples and must reconcile their disagreements at render time.

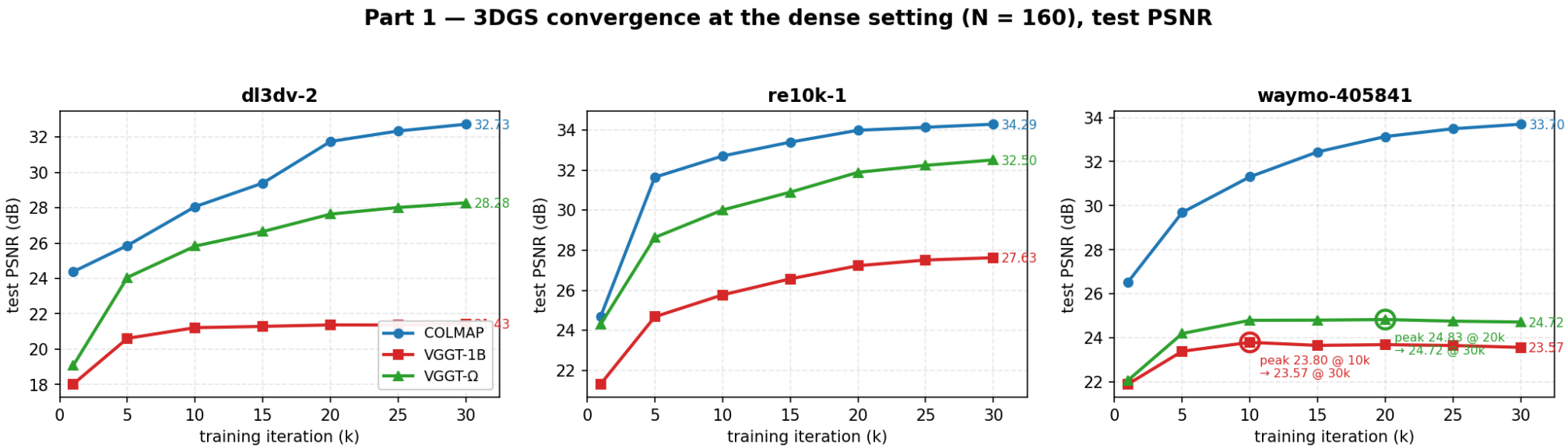
— The reported 0.000 px reprojection error is vacuous: a point seen once has no cross-view residual.

> **The contradiction, at 50 views (Re10k-1):**
> VGGT-Ω has **5× COLMAP's pose accuracy** (ATE 0.220 % vs 1.065 %) and still renders **1.35 dB worse**.

SO WHAT SHOULD YOU CONCLUDE?

- Pose accuracy alone cannot explain rendering quality.
- The feed-forward models are not worse reconstructors — they deliver competitive poses and dense but unconstrained geometry.
- Follow-up we did not run: keep VGGT's poses, re-triangulate points from multi-view correspondences.

Convergence: 30 k is the right budget at 160 views



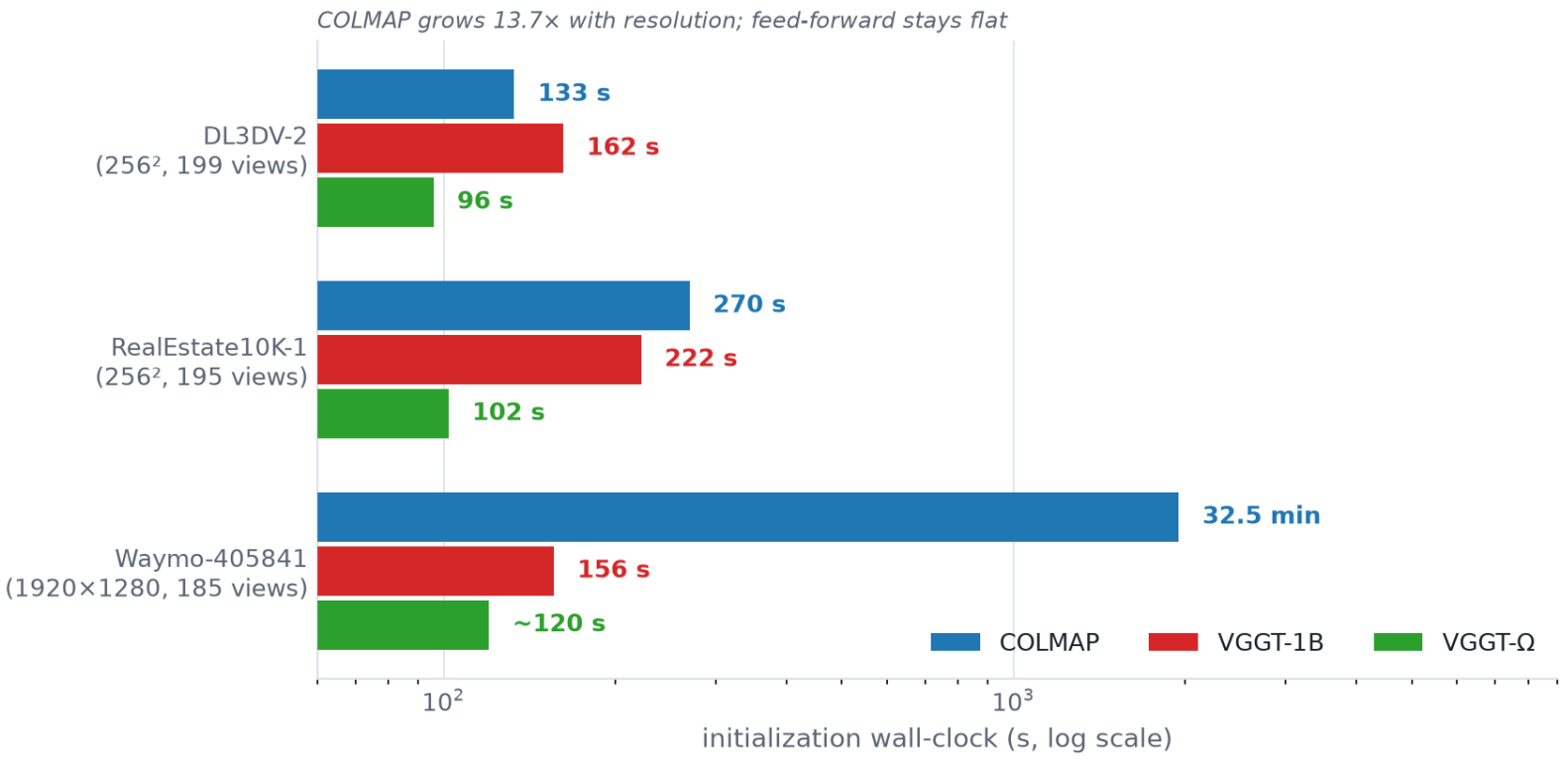
Hollow circles mark configurations whose peak occurs before 30 k and then decays; the two numbers give peak and final test PSNR.

7 of 9 configurations are still improving at 30 k — and all three COLMAP curves are.

Waymo is the exception: both feed-forward models peak early (10 k, 20 k) and then decay; COLMAP does not.

The "peak at 1 k" from the N=50 run does not occur at 160 — it was under-training, not over-fitting.

Speed and run-to-run stability



READING THE CHART

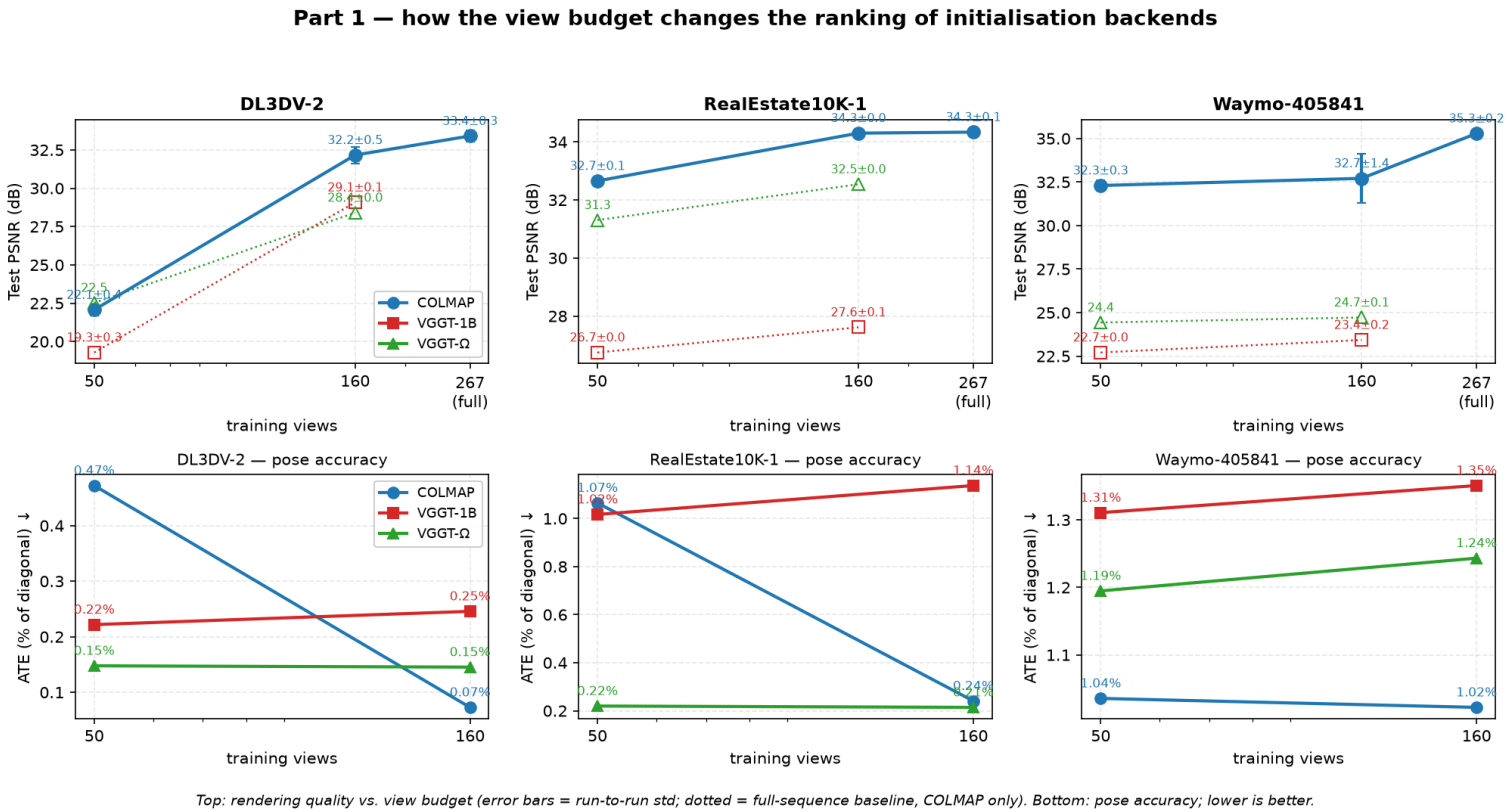
— Feed-forward: 70–220 s regardless of input resolution (every input is normalized to a fixed 512² window).

— COLMAP: 133 s → 1 950 s as resolution grows from 256² to 1920×1280.

AND IT REPRODUCES

- **8 of 9 configurations** reproduced to within 0.12 dB on a second GPU, rebuilt from scratch.
- **The one outlier:** VGGT-1B's exporter subsamples the cloud with an unseeded np.random.choice.

The gap widens as observations accumulate



Top: rendering quality vs. view budget (error bars = run-to-run std; dotted = full-sequence baseline, COLMAP only). Bottom: pose accuracy; lower is better.

COLMAP's pose error collapses 6.5× (DL3DV-2) and 4.4× (Re10k-1) from 50 → 160 views; feed-forward error does not move. VGGT-Ω gains the least (+5.85 dB vs COLMAP's +10.03 on DL3DV-2), which is what moves it from first place at 50 views to last at 160.

Seven silent failure modes

Every number above required eliminating these first. None produced an error message. Two individually exceeded the effects we report as findings.

#	Failure	Silent symptom	Impact
1	Diverged COLMAP camera	camera centre at 7.2×10^{9}; scene box 7.2×10^{9}	**+13.0 dB**
2	Double-normalized focal in the Ω export	exported fx = 0.4 instead of 204	**+13.4 dB**
3	Mixed alignment procedures	rotation error inflated 20× on a near-planar path	would mis-rank
4	Wrong Waymo GT convention	rotation error 23.5° instead of 1.38°	poses look broken
5	Dataset poses hide SfM cost	initialization time = 0	speed axis impossible
6	Missing sparse/0 directory	"Could not recognize scene type!"	pipeline won't start
7	LPIPS weights fetched at eval time	528 MB at 150 KB/s	~1 h of stalls

Failures 1 and 2 are the same species: scale errors that leave the optimizer running happily, detectable only by checking an intermediate quantity against a known reference. Watching the loss curve would not have caught either.

Takeaways

> **COLMAP-initialized 3DGS renders best on all three scenes, and the margin widens as observations accumulate.**

- **Pose accuracy is not the explanation.** COLMAP's error falls 4–6× from 50 → 160 views while feed-forward error stays flat — and at 50 views COLMAP has the worst poses yet the best renders.
- **VGGT-Ω is the best feed-forward option.** By far the most reproducible (std 0.03–0.07 dB), and it beats COLMAP at 50 views on DL3DV-2 — but it gains least from more views and falls to last place.
- **The decisive variable is point-cloud topology.** Track length 8.19 for triangulation vs 1.000 for per-frame depth unprojection.
- **What data-driven scaling buys:** a 70–100 s, resolution-insensitive, highly reproducible initializer — at the cost of 3–9 dB of fidelity.

Limitations: VGGT-1B's export is stochastic (±several dB); Mip-NeRF 360 is single-seed; no hybrid VGGT-poses + re-triangulated-points run; COLMAP timings are CPU-bound.